Unveiling Audience Preferences: A Big Data Analysis of IMDb Movie Ratings and Trends

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***Abstract*—Audience sentiment is an important driver in the entertainment business, but traditional methods of survey-based research are generally not scalable or objective. Big data analytics have changed this picture, and movie ratings available from the public serve as a good proxy for knowing audience sentiment and engagement. Textual reviews, however, are missing in most datasets, and this restricts traditional sentiment analysis. This work discusses almost 1 million of IMDb movie ratings within 17 genres based on rating-based sentiment classification. Our results show a decreasing trend in ratings over time, different sentiment distributions across genres, and prominent engagement patterns. These findings show the potential of organized big data in supporting decision-making for content creators, marketers, and streaming platforms.**

***Index Terms*—IMDb Ratings, Big Data Analytics, Sentiment Analysis, Audience Sentiment Analysis, Entertainment Industry Insights**

I. Introduction

The Entertainment sector is a fluid market where changing audience tastes are the driving force. Knowing what audiences prefer and what they dislike is essential for producers, production houses, and over-the-top players to make effective decisions about content development, promotion, and distribution. Conventionally, audience sentiment was measured through surveys, focus groups, and box office collections, but these tools are always limited in scope, prone to response errors, and do not allow for real-time course corrections. With the arrival of digital platforms, user-generated content, and web rating systems, the access to large-scale data has revolutionized the process of analyzing audience sentiment.

Big Data Analytics has proven to be a robust method to reveal in-depth insights into audience behavior [3] [4] through the use of enormous datasets, especially from online platforms where users review and rate films. One of the most exhaustive sources of such information is the Internet Movie Database (IMDb), a popular aggregation site that compiles audience ratings, metadata, and engagement metrics. IMDb offers organized data, such as movie genres, release years, age ratings, runtime, and more importantly, numeric ratings by millions of users across the globe. The capacity to probe this large data set allows one to conduct evidence-based investigations of audience taste, genre trends, and the temporal structure of movie ratings. Although text reviews carry significant sentiment-related insights, they are challenging to analyze owing to their linguistic nature and subjectivity. Further, there are users who do not offer text reviews. Hence, text-based sentiment analysis is not a comprehensive picture. To address this constraint, rating-based sentiment classification is employed in order to infer audience sentiment from numerical ratings and participation patterns. This enables a systematic study of high-level tendencies, including changes in popularity of genres, seasonally varying audience interest, and effects of demographic controls on participation levels.

This paper seeks to address important questions about audience behavior through a Big Data approach:
1) What genres are preferred by audiences and least preferred based on ratings?
2) How do audience preferences change over time?
3) How do demographic variables such as movie age ratings and runtime affect audience engagement?

4) What are the correlations between the most important measures of audience participation such as number of reviewers, reviews, and IMDb scores?
This study applies data mining, statistical analysis, and visualization to uncover audience preference, providing insight for filmmaking, marketing, and recommendation systems [1].

II. Related Work

A number of studies have investigated the taste of the audience in the entertainment sector through big data analytics, applying methods such as recommendation systems, sentiment analysis, and metadata-based audience modeling. Such methods are useful but also have some limitations, which this research seeks to overcome.

A. Sentiment Analysis in Movie reviews

Much of the audience preference analysis research has centered on sentiment analysis based on Natural Language Processing (NLP) methods. Researchers have used textual feedback from sites such as IMDb, Twitter, and Rotten Tomatoes to obtain the polarity of the sentiment, the emotional tone, and the impression of the viewer of films [1] [2]. Although these methods provide fine-grained insight into audience opinion, they tend to be limited by data availability, since not everyone provides written feedback. Moreover, language differences, sarcasm, and cultural variations make it difficult to interpret sentiments from text data accurately. In light of these constraints, other approaches, like rating-based sentiment classification, offer a more formal and scalable way of analyzing audience sentiment [8].

Additional studies have put algorithms originating from machine learning and deep learning approaches to use for IMDb sentiment analysis and prediction. The studies above, i.e., [9], [10], [12], [13], [15], and [19], utilize a number of different models including classifiers, hybrid models, etc. Also, a more unique approach as stated in [14], employed social media markers (i.e., Facebook) to determine film success and audience sentiment.

B. Movie Recommendation Systems

This is another significant field of research that includes movie recommendation systems that forecast audience tastes through collaborative filtering and machine learning models [3]. These systems take advantage of user behavior patterns, historical interactions, and similarity-based filtering to suggest movies based on individual preferences. Though efficient, such models are based on a lot of user-specific data and lack the capability to capture overall industry-wide patterns. Our research differs from these methods by moving the focus away from individualized recommendations to a macro-level examination of audience tastes based on collective IMDb rating data.

C. Big data in Entertainment Analytics

Application of big data methods in investigating audience behavior and film performance has been heavily discussed in existing literature. The analysis of IMDb metadata such as ratings, duration, genre, year of release, and demographics has been utilized to measure viewership trends and audience interaction [5]. Researches highlight the role of the structured film data in predictive strength of trends within the popularity of films, without having an all-inclusive sentiment classification model free of textual processing [6] [7].

D. Comparison with Existing methods

While the majority of current sentiment analysis papers have relied on Natural Language Processing (NLP) that has utilized deep learning methods such as CNN, LSTM, and BERT [3] [11] [18], we have taken a different approach by exclusively addressing the rating-based sentiment classification. It completely eliminates the hazards associated with textual ambiguity, sarcasm, or multi-lingual interpretation, which are normally inherent to text-based models. Additionally, while most of the traditional film recommendation systems are centered on collaborative filtering and user-based activity [2], these models do not necessarily capture the audience sentiment at the macro-level. Our model on the other hand uses structured IMDb meta data, e.g., numeric ratings, genres, runtime, and reviewer counts, to derive high-level viewership trends and genre participation. Thus, creating a model that is more scalable and objective, and provides content producers and platforms actionable insight without user-text input.

This study presents a new method of analyzing audience sentiment by employing IMDb ratings rather than tracking the text reviews. It methodologically integrates various analysis method to:

- Sentiment classification based on IMDb ratings
- Tracking shifts in genre preferences over time
- Establishing relationships between engagement measures including reviewer count and ratings.

Utilizing big data, it offers clear and useful perspectives of audience preferences, which will be useful for content creators, marketers, and the entertainment industry.

III. Methodology

The following study applies a formal procedure includes four major phases: Data Preprocessing, Sentiment Analysis, Correlation Analysis, and Evaluation of Trends for an Audience with IMDb Film Scores as Ratings.

A. Dataset & Simulation Overview

The data used in this research is drawn from Kaggle's IMDb Movie Reviews dataset [20], which has around 1 million distinct movie ratings across 17 genres - Action, Adventure, Animation, Biography, Comedy, Crime, Drama, Fantasy, History, Horror, Music, Mystery, Romance, Sci-Fi, Sport, Thriller and War. Each movie record has the following features of metadata:
1) Date of release of the movie
2) Run length
3) IMDb rating

4) Movie rating (PG-13, R, etc)
5) Number of IMDb raters and
6) Number of reviews per movie

- Software used: Python, main libraries were NumPy, Pandas and Seaborn
- No train-test split was performed

B. Proposed Framework

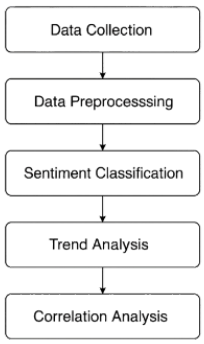

Fig. 1. Analytical Framework

C. Data Preprocessing

Dataset is preprocessed to make it clean and structured for analysis, these include:

- Merging Multiple Genre Files: Because the dataset is composed of different files for every genre, all files were merged into one dataframe so that cross-genre comparisons could be easily made.
- Feature Engineering: The runtime feature, as initially recorded in hours and minutes (hh:mm), was reformatted to total minutes for uniform numerical analysis.
- Missing Data Management: Records with missing IMDb ratings, number of raters, or reviews were excluded to preserve dataset integrity and prevent biases in trend analysis.

These preprocessing operations provided a uniform dataset, permitting precise audience preference estimations.

D. Sentiment Classification

Since in our dataset, there were no textual reviews available, a rating-based sentimation was used to derive the audience analysis. This was its mathematical representation:

$$S(r) = \begin{cases} \text{Positive} & r \geq 7.0 \\ \text{Neutral} & 5.0 \leq r < 7.0 \\ \text{Negative} & r < 5.0 \end{cases}$$

E. Audience Trend Analysis

For getting deeper insights, audience trends were analyzed across various dimensions:

1) Genre-Based Trends: The highest rated and most reviewed genres were determined to identify the preferences of the audience. A box plot analysis was performed to analyze the distribution of ratings by genre, focusing on differences in audience perception.

2) Time-based Trends: A time-series analysis was used to analyze the change in IMDb ratings over time, determining changes in audience preferences. Trends were analyzed over various decades to evaluate changes in movie ratings and viewing patterns.

3) Demographic-based Trends: The effect of film age ratings (e.g., PG-13, R, G) on viewer sentiment was examined to determine to what extent content limitation affects viewer opinion. The correlation between runtime and IMDb ratings was measured to establish if longer films have higher or lower viewer ratings.

F. Correlation Analysis

To determine the interdependencies among the core engagement metrics, a correlation heatmap was created to investigate the relationships among:

- IMDb ratings
- Number of reviewers
- Number of reviews

This analysis assists in determining whether an increased number of raters provides more consistent ratings and if highly rated movies (movies with large review numbers) will have higher or lower audience ratings.

Through data preprocessing, rating-based sentiment analysis, audience trend analysis, and correlation studies, this approach offers a holistic and data-centric understanding of audience sentiments on IMDb. The findings of this framework have important implications for the entertainment sector, helping filmmakers, producers, and streaming platforms make informed decisions about content creation and marketing strategies.

IV. Results

In this section, we will mainly discuss the key findings of our research that are primarily focused on sentiment analysis, genre analysis, time analysis, demographic analysis and correlation analysis.

A. Sentiment Analysis

The sentiment classification based on ratings showed clear patterns of audience sentiment:

- 76.76% of films were positively rated (IMDb $\geq$ 7.0), which reflects a overall audience preference for highly rated movies.
- 22.76% of films were neutral (IMDb 5.0 – 6.9), indicating that a considerable number of films are given mixed reviews.
- 0.47% of films were unfavorably rated (IMDb $<$ 5.0), indicative of high viewer dissatisfaction with particular movies [16].

These findings show that audiences will favorably rate films, although there are a minority of films that are poorly received.

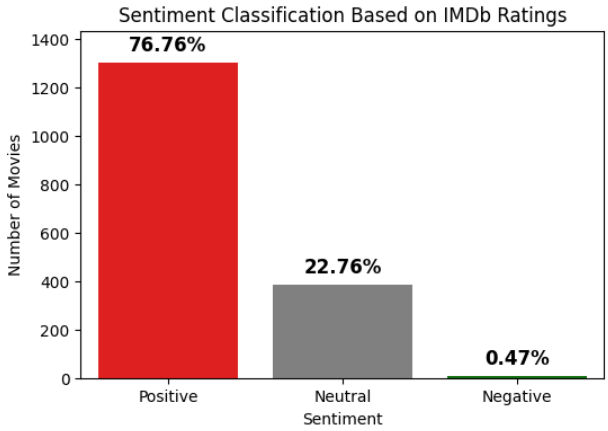

Fig. 2. Sentiment Analysis

B. Genre-Based Insights

Evaluating IMDb scores for various genres offered insightful evidence about what attracts viewers:

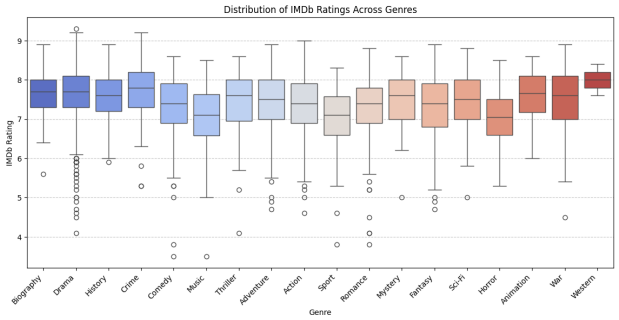

Fig. 3. Genre-Based Analysis

- Highest average IMDb ratings belonged to the Western, War and Animation genres, which indicate people like movies featuring strong storylines, real-story plots, and since Western genre has a very high median rating reflecting strong widespread audience preference.
- Horror, Comedy, and Drama are highly variable in ratings, so audience views are more polarized. These genres possess long whiskers and many outliers, indicating a combination of highly-rated and low-rated movies.
- Horror and Romance films seem to have the lowest median ratings [17].
- Biography, History, and Crime films have higher median ratings with comparatively smaller spreads, indicating they are uniformly well-rated. These genres appear to be more dependable in terms of quality.
- Fantasy, Romance, and Horror have a lower median with a wide spread, reflecting a combination of audience opinions.
- There are a number of low-rated outliers in Drama and Horror, perhaps because of extremely poorly received films in these genres. A few Western and Animation films also have high outliers, reflecting a few highly rated films.

C. Time-Based Trends

The research also explored how audience tastes changed over time:

1) Average IMDb Rating over years:

- There was a declining trend in IMDb ratings, indicating that audiences have become increasingly critical over the years. This may be due to reasons like increased expectations from contemporary films, greater exposure to varied content, and shifting trends in cinema.

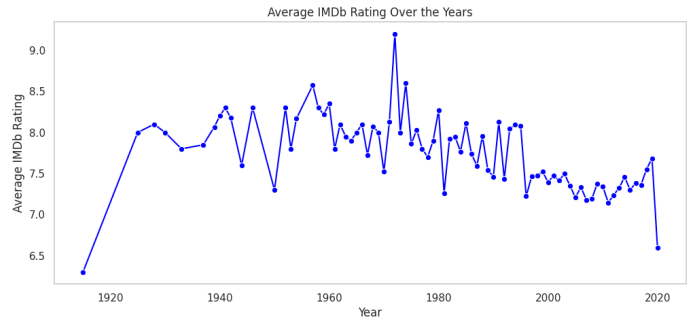

Fig. 4. Average IMDb Ratings over the years

2) Total Number of Reviews over the years:

- Between 1920 and approximately 1980, the number of reviews was extremely low with little activity.
- Following the 1980s, there was a slow increase in aggregate reviews. A steep rise is seen after the year 2000, showing a sudden upsurge in viewership activity towards films.
- The spike around 2020 indicates a huge increase in online consumer reviews, possibly because of the popularity of streaming services and social media impact.
- The dip towards the end may be because of missing data for the most recent year or a fall in film releases (e.g., because of COVID-19 during 2020).

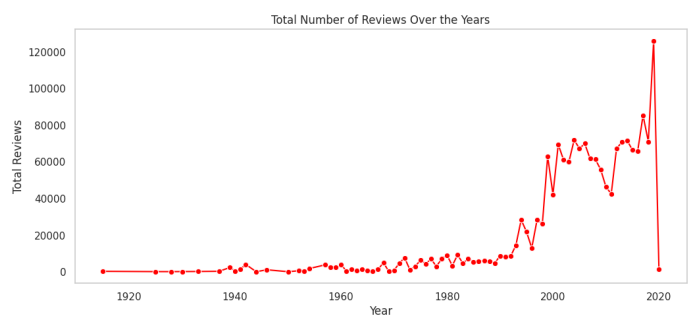

Fig. 5. Number of Reviews over the years

3) Total Number of Raters over the years:

- Between 1920 and 1960, very little was done in the way of user ratings. The size of raters began growing after 1960, indicating the rising impact of films.
- A considerable spike post-1990s indicates the appearance of IMDb and similar rating websites, where people had

more avenues to contribute. The peak of 2000-2015 is the best time of audience engagement, when millions of users actively rated films.

- A declining trend after 2015-2020 indicates shifting audience behavior—maybe fewer individuals are rating films, moving towards other engagement such as social media chatter and video content.

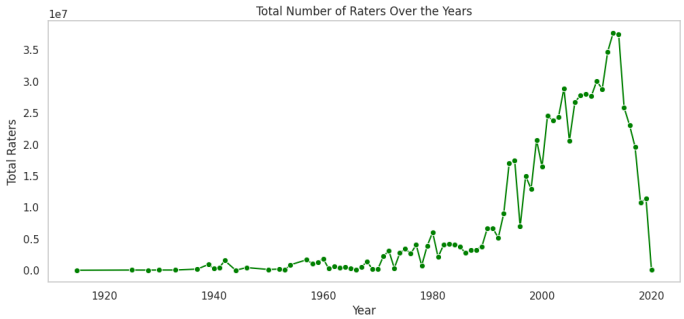

Fig. 6. Number of Raters over the years

These findings reflect a change in audience attitude, highlighting the necessity for modern filmmakers to be responsive to changing viewer demands.

D. Demographic Impact

Several demographics were examined to determine their effect on IMDb scores:

1) Average IMDb Rating by Movie Rating Category:

- The highest IMDb rating is of the ”NULL” category, perhaps suggesting films with missing or unclassified ratings are still highly rated.
- ”Passed” and ”Approved” categories (earlier classifications) also possess good ratings, which suggests classic films are popular.
- ”TV-MA” (Mature Audiences) and ”TV-PG” have lower average ratings, suggesting that audiences may be more discerning of television programming.
- ”R-rated” movies (Restricted) are not the best-rated, perhaps because of the mixed nature of films in this category.

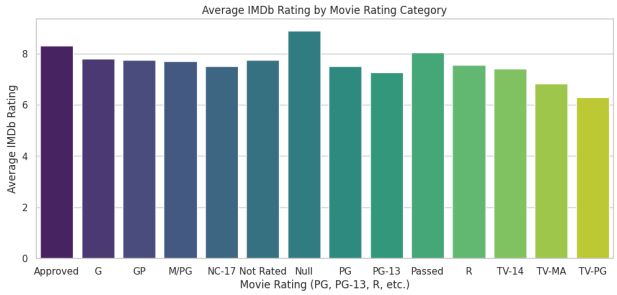

Fig. 7. Movie age ratings

2) Total Reviews by Movie Rating Category:

- ”PG-13” and ”R” films lead the pack in the number of reviews, indicating that these categories have the most attention from viewers. This is also because most of the movies are successful blockbusters and mainstream releases.
- ”Approved,” ”G,” and ”Not Rated” categories contain few reviews, which means that family-friendly or older movies might not be discussed as much online.
- ”NC-17” (Adults Only) films also contain very low reviews, which is probably because they have limited releases and limited audiences.

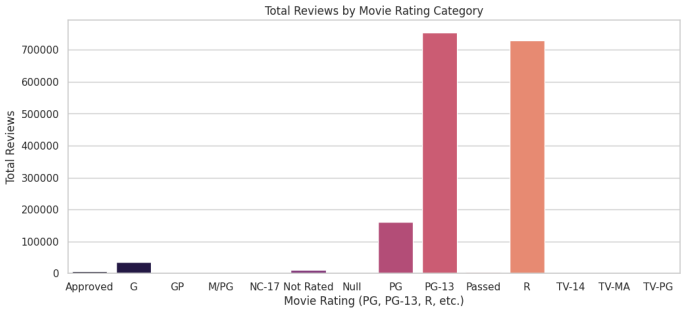

Fig. 8. Total reviews by Movie Rating

3) Total Raters by Movie Rating Category:

- PG-13 and R-rated films have the most raters, exceeding 300 million. This means that the most significant group of moviegoers is in the age bracket that views PG-13 and R-rated films.
- G-rated films have far fewer raters than PG-13 and R. This implies that films for children might not be as much of an online draw or source of discussion.
- NC-17, Not Rated, and Null categories have very little activity. Films in these categories tend to have limited releases, less visibility, or appeal to niche groups. The ”Not Rated” category can consist of older movies that were not rated under contemporary rating systems.

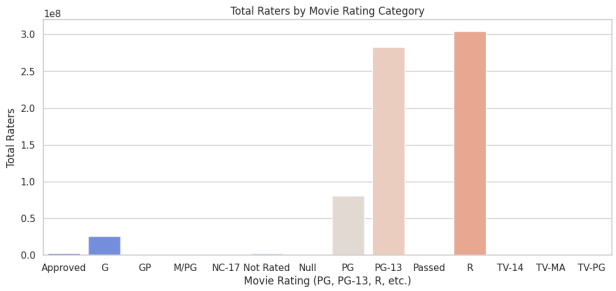

Fig. 9. Total Raters by Movie Rating

E. Correlation Analysis

The correlation between core audience engagement metrics [18] was tested using correlation analysis:

1) IMDb rating vs Other factors:

- A moderate positive correlation (0.57) means that films with more raters have higher ratings. But it is not a very strong correlation, so high ratings are not strictly based on the number of raters.
- A weak positive correlation (0.27) implies that films with more reviews tend to have slightly higher ratings, but it is not a strong predictor.
- A weak positive correlation (0.28) indicates that longer films may get a bit higher ratings, but rating is not influenced much by duration.

2) Number of Raters vs Other factors:

- Higher numbers of raters tend to increase the number of reviews (0.60), reflecting high audience engagement for popular movies.
- Weak correlation (0.27) indicates that film length has minimal effect on the number of individuals rating it.

3) Number of Reviews vs Run Length:

- Slight positive correlation (0.30) indicates that longer films receive more reviews, perhaps because of more lengthy explanations or franchise-film movies creating buzz.

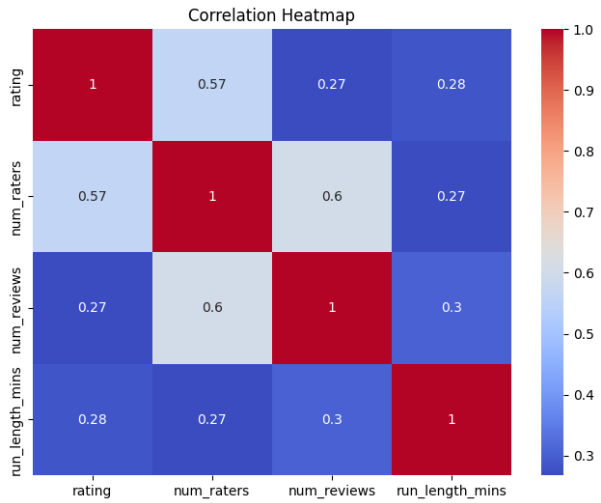

Fig. 10. Correlation Analysis

F. Comparative Insights and Key Findings

The comparative overview of the current state of sentiment analysis techniques and our rating-based approach revealed several important findings. First, our methodology is not dependent on a language model based on textual data and plenty of Natural Language Processing (NLP) processing, while other sentiment analysis methodologies were dependent on textual data and had extensive pre-processing requirements. We primarily relied on structured IMDb metadata to determine sentiment, a method that could be scaled and is language independent. The data revealed observable genre preference trends, such as the median ratings in war, biography, and animated films were higher than many other genres, and it demonstrated declining trends in ratings over time - these trends are not consistently identifiable in NLP systems. Other audience engagement measures in our analysis, including the total number of reviews and raters, were strongly correlated to sentiment metrics and validate the potential for determining audience preferences based on numerical metadata. Overall, our framework yields an efficient and lightweight substitute to consistently track content preferences and audience engagement in large entertainment datasets.

V. Conclusion and Future Scope

Through the use of an IMDb dataset, we propose a deep dive into audience participatory behavior, sentiment mappings, and genre preferences through a rating-based sentiment classification and big data techniques in our work. We found clear audience dispositions of specific genres (War, Animation, Biography) as well as a general down-trend of IMDb scores over time indicating an increasingly discriminating audience behavior over time.

On the one hand, correlation analysis supported the notion of a moderate relationship between having more unique reviewers would yield a better IMDb rating, but on the other, audiences' participatory behavior would differ across demographic variables such as rating and movie run time.

Different than previous literature that mainly focused on textual based sentiment analysis, our methodology is language independent and does not rely on text, rather only structured metadata to analyze audience behavior. Future research could expand on this work with the inclusion of Natural Language Processing (NLP) to add text-on-text sentiment enrichment, machine learning models (regression or neural networks) to predict audience rated status or box-office success, and geographic and/or cultural distinctions in sentiment mappings.

In addition, the changing role of streaming services, seasonal popularity of genres, and the audience 'real-time' feedback would all add additional richness to our understanding of digital audiences within the entertainment industry.

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